Status	Finished
Started	Saturday, 24 October 2026, 9:47 PM
Completed	Saturday, 24 October 2026, 9:49 PM
Duration	2 mins 23 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**
Complete
Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

- a. Linear Search
- b. Bubble Sort
- c. Merge Sort
- d. Binary Search

The correct answer is: Merge Sort

Question **2**
Complete
Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

- a. Bubbling up the smallest element
- b. Sorting the entire list in one pass
- c. Dividing the list into halves
- d. Bubbling up the largest element to its correct position

The correct answer is: Bubbling up the largest element to its correct position

Question **3**
Complete
Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

- a. Removing duplicates from a list
- b. Arranging data in a specific order
- c. Finding a specific element in a list
- d. Merging two datasets

The correct answer is: Arranging data in a specific order

Question **4**
Complete
Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

- a. Sorting the entire list sequentially
- b. Dividing the input into smaller subproblems
- c. Comparing each element with the others
- d. Combining sorted sublists

The correct answer is: Dividing the input into smaller subproblems

Question **5**
Complete
Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

- a. arrange()
- b. sorted()
- c. order()
- d. sort()

The correct answer is: sort()

Question **6**
Complete
Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

- a. Merge Sort
- b. Linear Search
- c. Binary Search
- d. Bubble Sort

The correct answer is: Bubble Sort

Question **7**
Complete
Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

- a. Merge Sort
- b. Heap Sort
- c. Quick Sort
- d. Bubble Sort

The correct answer is: Bubble Sort

Question **8**
Complete
Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

a. Bubble Sort cannot handle duplicate elements
b. Bubble Sort is difficult to implement
c. Bubble Sort does not guarantee sorted order
d. Bubble Sort is less efficient for large lists

The correct answer is: Bubble Sort is less efficient for large lists

Question **9**
Complete
Mark 1.00 out of 1.00

What is sorting in the context of computer science?

a. Deleting data from a list
b. Inserting data into a list
c. Searching for data in a list
d. Arranging data in a particular format

The correct answer is: Arranging data in a particular format

Question **10**
Complete
Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

a. Merge Sort
b. Insertion Sort
c. Quick Sort
d. Bubble Sort

The correct answer is: Merge Sort

Question **11**
Complete
Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

a. Distribution
b. Insertion
c. Selection
d. Extraction

The correct answer is:
Insertion

Question **12**
Complete
Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

- a. Sorting data sequentially
- b. Sorting data in a single pass
- c. Rearranging data without sorting
- d. Dividing the input into parts, solving each part, and combining the solutions

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **13**
Complete
Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 200,470,80,150,40,90,300,400,70,120
- b. 80,150,200,470,40,90,300,400,70,120
- c. 40,70,80,90,120,150,200,300,400,470
- d. 40,80,90,150,200,300,400,470,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **14**
Complete
Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

- a. Merge Sort
- b. Bubble Sort
- c. Binary Search
- d. Linear Search

The correct answer is: Merge Sort

Question **15**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

- a. It compares adjacent elements and swaps them if necessary
- b. It repeatedly finds the minimum element and moves it to the sorted part of the list
- c. It builds a sorted array one element at a time
- d. It divides the list into two halves, sorts each half, and then merges them

The correct answer is: It divides the list into two halves, sorts each half, and then merges them